In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

ALL INDIA TEST SERIES

FIITJEE **JEE (Advanced), 2017**

CONCEPT RECAPITULATION TEST – I

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 186**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 05)** contains 5 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (06 to 13) contains 8 multiple choice questions which have one or more than one correct answer. Each question carries **+4 marks** for correct answer and **– 2 mark** for wrong answer.

(ii) **Section-C (01 to 05)** contains 5 Numerical based questions with single digit integer as answer, ranging from 0 to 9. Each question carries **+3 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

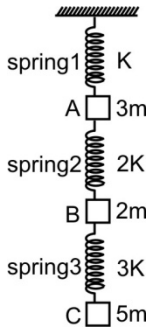
PART – I

SECTION – A

Straight Objective Type

This section contains **5 multiple choice questions** numbered 1 to 5. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

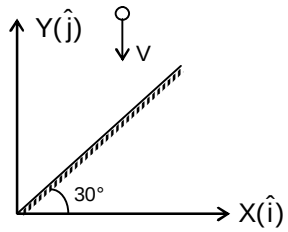
1. The system of springs and masses are in equilibrium. If the spring 2 is cut then the initial acceleration of block B will be.



(A) $\frac{7}{2}g$ downward (B) 2g downward

(C) 2g upward (D) $\frac{7}{2}g$ upward

2. An object moves with velocity $V(-\hat{j})$ towards the inclined plane mirror as shown in figure. The velocity of the image is



(A) $V\hat{j}$ (B) $V\frac{\sqrt{3}}{2}(-\hat{i})+\frac{V}{2}\hat{j}$

(C) $V\frac{\sqrt{3}}{2}\hat{i}+\frac{V}{2}\hat{j}$ (D) $\frac{V}{2}(-\hat{i})+V\frac{\sqrt{3}}{2}(\hat{j})$

3. When a beam of accelerated electrons hits a target, a continuous X-ray spectrum is emitted from the target. Which of the following wavelength is absent in the X-ray spectrum, if the X-ray tube is operating at 40.000 volts?

(A) 0.25 Å (B) 0.5 Å

(C) 1.5 Å (D) 1.0 Å

Space for Rough work

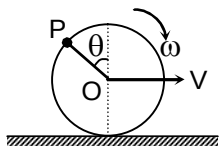
4. The electric potential between a proton and an electron is given by $V = V_0 \ln\frac{r}{r_0}$, where r_0 is a constant. Assuming Bohr's model to be applicable, write variation of r_n with n, n being the principal quantum number.
(A) $r_n \propto n$ (B) $r_n \propto 1/n$
(C) $r_n \propto n^2$ (D) $r_n \propto 1/n^2$

5. A string of mass m is fixed at both ends. The fundamental tone oscillation are excited in the string with angular frequency ω and maximum displacement amplitude A. Find the total energy contained in the string.
(A) $\frac{1}{2}m\omega^2A^2$ (B) $\frac{1}{4}m\omega^2A^2$
(C) $\frac{1}{6}m\omega^2A^2$ (D) $\frac{1}{8}m\omega^2A^2$

Multiple Correct Choice Type

This section contains **8 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

6. A disc of radius *R* rolls on a horizontal surface with linear velocity *V* and angular velocity ω. There is a point *P* on circumference of disc at angle θ with upward vertical diameter measured anticlockwise see figure, which has a vertical velocity. Here θ is equal to



(A) $\pi + \sin^{-1}\frac{V}{R\omega}$ (B) $\frac{\pi}{2} - \sin^{-1}\frac{V}{R\omega}$
(C) $\pi - \cos^{-1}\frac{V}{R\omega}$ (D) $\pi + \cos^{-1}\frac{V}{R\omega}$

Space for Rough work

7. A point moves with deceleration along the circle of radius R so that at any time its tangential and normal acceleration are equal in magnitude. At the initial moment t =0, the velocity of the point is v_0. The velocity of the point will be

(A) $v = \dfrac{v_0}{1+\dfrac{v_0 t}{R}}$ at t second

(B) $v = v_0 e^{-S/R}$ after displacements

(C) $v = v_0 e^{-SR}$ after displacements

(D) $v = \dfrac{v_0}{1-\dfrac{v_0 t}{R}}$ at t second

8. A block of mass 1 kg moves under the influence of external forces on a rough horizontal surface. At some instant, it has a speed of 1 m/s due east and an acceleration of 1 m/s^2 due north. The force of friction acting on it is F
(A) F acts due west
(B) F acts due south
(C) F acts in the south west direction
(D) the magnitude of F cannot be found from the given data

9. A particle is taken from point A to point B under the influence of a force field. Now it is taken back from B to A and it is observed that the work done in taking the particle from A to B is not equal to the work done in taking it from B to A. If W_{nc} and W_c is the work done by non conservative forces and conservative forces present in the system respectively, ΔU is the change in potential energy, Δ k is the change is kinetic energy, then

(A) $W_{nc} - \Delta U = \Delta k$

(B) $W_c = -\Delta U$

(C) $W_{nc} + W_c = \Delta k$

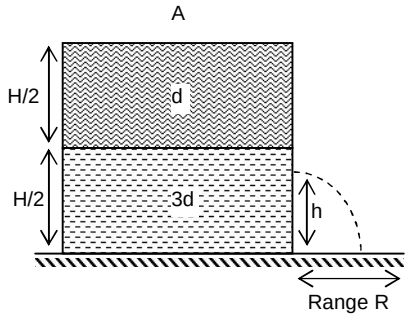
(D) $W_{nc} - \Delta U = -\Delta k$

10. A horizontal disc rotates freely about a vertical axis through its centre. A ring, having the same mass and radius as the disc, is now gently placed on the disc. After some time, they rotate with a common angular speed.
(A) some friction exists between the disc and the ring.
(B) the angular momentum of the disc plus ring is conserved.
(C) the final common angular speed is $\dfrac{2}{3}$ rd of the initial angular velocity of the disc.
(D) $\dfrac{2}{3}$ rd of the initial kinetic energy changes to heat.

Space for Rough work

11. If electron of the hydrogen atom is replaced by another particle of same charge but of double the mass, then:
(A) Bohr radius will increase
(B) ionisation energy of the atom will be doubled
(C) speed of the new particle in a given state will be lesser than the electron's speed in same orbit
(D) gap between energy levels will now be doubled

12. A container of large uniform cross sectional area A resting on a horizontal surface holds two immiscible non viscous and incompressible liquids of density d and 3d, each of height H/2. The lower density liquid is open to the atmosphere having pressure P_0. A tiny hole of area a(a<<A) is punched to the vertical side of lower container at a height h (0<h<H/2) for which range is maximum.



(A) h = H /3
(B) Range R = $\frac{2H}{3}$
(C) Range R = $\frac{3H}{2}$
(D) Velocity of efflux $v = \sqrt{\frac{2}{3}gH}$

13. When a monochromatic point source of light is at a distance of 0.2 m from a photoelectric cell, the cut off voltage and the saturation current are respectively 0.6V and 18.0 mA. If the same source is placed 0.6 m away from the photoelectric cell, then:
(A) the stopping potential will be 0.2 volt
(B) the stopping potential will be 0.6 volt
(C) the saturation current will be 6.0 mA
(D) the saturation current will be 2.0 mA

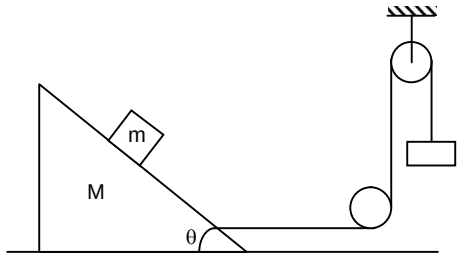
Space for Rough work

SECTION – C

(One Integer Value Correct Type)

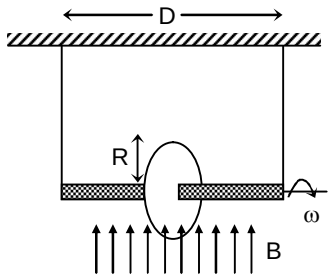
This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. Find the mass of the hanging block in kg, which will prevent the smaller block from slipping over the triangular block. All the surfaces are friction less and the strings and pulleys are light. (Given m = M = 1 kg, $\cot\theta = 2$)



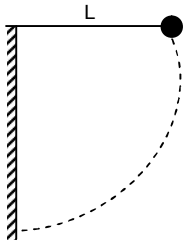
2. An ideal gas is expanding such that PT^2 = constant. The coefficient of volume expansion of the gas is A/T where T is temperature in kelvin. Find the value of A.

3. The axle of a circular wheel of radius R is held horizontally by two identical strings of equal lengths separated by a distance D. The tension in each string is T_0. The rim of the wheel carries a total charge + Q distributed uniformly on it. The wheel is vertical and is kept in a uniform vertical magnetic field $\vec{B}$. It is now rotated at an angular speed ω. The string break at a tension of $3T_0/2$. If the maximum possible value of ω at which the wheel can be rotated without breaking a string is $\frac{KT_0D}{QR^2B}$. Then find the value of K.

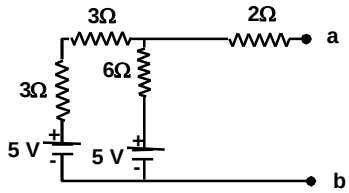


4. A simple pendulum is suspended from a peg on a vertical wall. The pendulum is pulled away from the wall to a horizontal position and released. The ball hits the wall, the coefficient of restitution is $\frac{2}{\sqrt{5}}$.

   What is the minimum number of collisions after which the amplitude of oscillations becomes less than 60 degrees ?



5. Determine the resistance in ohm is to be connected across a, b in given circuit to dissipate maximum power through it.



Space for Rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

This section contains **5 multiple choice questions** numbered 1 to 5. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

1. 100 mL of a sample of hard water requires 25.1 mL of 0.02 N H_2SO_4 for complete reaction. The hardness of the sample is:
[Density of the sample = 1 g/mL]
(A) 200 ppm (B) 250 ppm
(C) 251 ppm (D) 258 ppm

2. Which of the following is not a property of NaOH?
(A) It is hygroscopic in nature (B) It decolourizes chlorine gas
(C) It behaves as a bleaching agent (D) Heat is liberated when it reacts with HCl

3. For the gaseous reaction, $A(g) \rightarrow 4B(g) + 3C(g)$ is found to be first order with respect to A. If at the starting the total pressure was 100 mm Hg and after 20 minutes it is found to be 400 mm Hg. The rate constant of the reaction is
(A) 20 min^{-1} (B) 1.2×10^{-3} sec^{-1}
(C) 5.7×10^{-4} sec^{-1} (D) 3.4×10^{-2} sec^{-1}

4. Four separate solutions (each of 0.1 M) of four sodium salts NaW, NaX, NaY and NaZ has pH 7.0, 9.0, 10.0 and 11.0 respectively. The acid with the lowest pKa value is
(A) HW (B) HX
(C) HY (D) HZ

5. What would be the formula of an ionic solid which crystals contains b.c.c unit cell in which X^{2+} ions are present at the body centre of Y^{2-} ions at the corners?
(A) X_2Y (B) XY
(C) XY_2 (D) X_2Y_3

Space for Rough work

Multiple Correct Choice Type

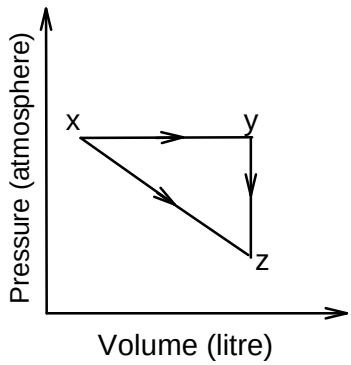
This section contains **8 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

6. In which of the following option(s) the dipole moment of the left side molecule is greater than that of the right side molecule?
(A) NH_3, NF_3 (B) SCl_2, $BeCl_2$
(C) SF_4, CF_4 (D) PF_5, IF_5

7. K_C for the reaction 2A (g) + B(g) $\rightleftharpoons$ A_2B(g) is 2×10^{-2} at 25^oC and it is 2×10^{-1} at 50^oC. Choose the incorrect statement (s) from the following
(A) Decomposition of A_2B is favoured at low temperature
(B) Formation of A_2B is favoured at low pressure.
(C) Formation of A_2B is favoured by introducing an inert gas at constant pressure.
(D) Decomposition of A_2B is favoured by removing B(g) from the reaction mixture

8. Which carbides react with water to give methane?
(A) CaC_2 (B) Be_2C
(C) Mg_2C_3 (D) Al_4C_3

9. For an ideal gas, consider only P – V work in going from an initial state X to final state Z. The final state Z can be reached by either of the two paths shown in figure. Which of the following choice (s) is (are) correct? [Take ΔS as change is entropy ΔH change is enthalpy and W as work done]



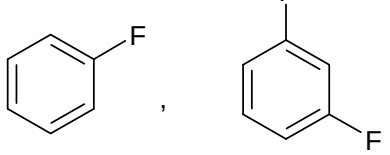
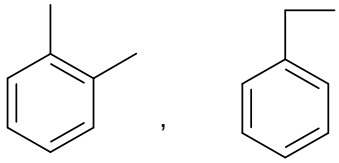
(A) $\Delta S_{x\to z} = \Delta S_{x\to y} + \Delta S_{y\to z}$
(B) $W_{x\to z} = W_{x\to y} + W_{y\to z}$
(C) $W_{x\to y\to z} = W_{x\to y}$
(D) $\Delta H_{x\to y\to z} = \Delta H_{x\to z}$

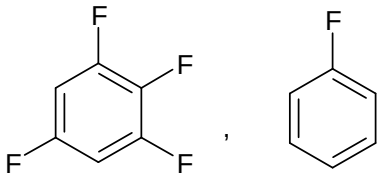
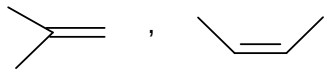
Space for Rough work

10. Which of the following ion(s) form(s) precipitate with aqueous ammonia solution in presence of ammonium chloride?
(A) Ca^{2+} (B) Fe^{3+}
(C) Cr^{3+} (D) Al^{3+}

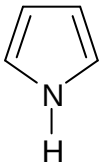
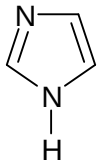
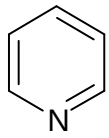
11. Which of the following ion(s) maintain an identical d-orbital configuration while forming octahedral complexes with strong field as well as weak field ligands according to crystal field theory?
(A) Fe^{2+} (B) Co^{2+}
(C) Ni^{2+} (D) Zn^{2+}

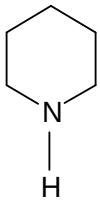
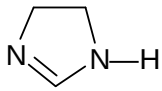
12. Which of the following pair has same net dipole moment

(A)  (B) 

(C)  (D) 

13. Arrange the following in decreasing order of basic strength?

(i)  (ii)  (iii) 

(iv)  (v) 

(A) iv > v > ii > iii > i (B) v > iv > ii > iii > i
(C) i > ii > iii > iv > v (D) v > iv > iii > ii > i

Space for Rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. What is the sum of azimuthal quantum number of all the valence electrons of aluminum? [At. No. of Al = 13]

2. How many S – O single bonds are present in Caro's acid (H_2SO_5)?

3. The formula of a spinel is $MgAl_2O_4$. How many effective no. of anionic vacancy for unit cell will be created if the oxide ions of the spinel are replaced by $X^{-8/3}$ ions?

4. The radiator of a car contains 50.5 g water. If 6.2 g of glycol ($HOCH_2CH_2OH$) is added to it, the solution freezes at -4^oC. How many gram of ice will be separated in the process? [K_f = 1.86 K kg mol^{-1}]

5. How many maximum gram of aluminium will be deposited by passing 96500 ampere current for one second through molten aluminium nitrate?

Space for Rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **5 multiple choice questions** numbered 1 to 5. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

1. If A and B are orthogonal square matrices of same order and $|A| + |B| = 0$, then the matrix $A + B$ is always
   (A) Invertible (B) Singular
   (C) Non – Null matrix (D) None of these

2. 3 Tangents are drawn at random to a given circle in a plane. The probability that the circle is inscribed in the triangle formed by them is
   (A) $\frac{1}{2}$ (B) $\frac{1}{3}$
   (C) $\frac{1}{4}$ (D) $\frac{1}{8}$

3. The number of 6 digit numbers which are divisible by 3 and do not contain the digit 3 is
   (A) $2^3 3^6$ (B) $2^3 3^9$
   (C) $2^4 3^6$ (D) $2^4 3^9$

4. A monic quadratic trinomial P (x) is such that $P(x) = 0$ and $P(P(P(x))) = 0$ have a common root, then
   (A) $P(0).P(1) > 0$ (B) $P(0)P(1) < 0$
   (C) $P(0)P(1) = 0$ (D) Data insufficient

5. The number of non – zero $z \in C$ satisfying $z^3 = 18 + 26i$, where $z = x + iy$ and x, y are integers, is $\left(i = \sqrt{-1}\right)$
   (A) 0 (B) 1
   (C) 2 (D) 3

Space for Rough work

Multiple Correct Choice Type

This section contains **8 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

6. If $f(x+y)=f(x)+f(y)+|x|y+xy^2 \forall x,y \in R$ and $f'(0)=0$, then
   (A) f is differentiable for all $x \in R$
   (B) f is non differentiable for at least one $x \in R$
   (C) f is twice differentiable for all $x \in R$
   (D) f is twice differentiable for all $x \in R-\{0\}$

7. Let $F:(1,\infty)\to R$ be a function defined as $F(x)=\int_x^{x^2}\frac{dt}{\ell n\, t}$. Which of the following statements are correct?
   (A) $F(x)$ is strictly decreasing
   (B) Range of $F(x)$ is $(0,\infty)$
   (C) Range of $F(x)$ is $(\ell n\, 2,\infty)$
   (D) $F(x)$ is one – one function

8. The sequence 1, 3, 4, 9, 10, 12, 13,……consists of all those positive integers which are powers of 3 or sums of distinct powers of 3. The hundredth term of this sequence is less than
   (A) 982
   (B) 983
   (C) 981
   (D) 980

9. A chord of negative slope from the point $P(\sqrt{264},0)$ is drawn to the ellipse $x^2+4y^2=16$. This chord intersects the ellipse at A and B (O is the origin).
   (A) The maximum area of ΔAOB is 4
   (B) The maximum area of ΔAOB is 8
   (C) If the area of ΔAOB is maximum, then the slope of line AB is $\frac{-1}{2\sqrt{2}}$
   (D) If the area of ΔAOB is maximum, then the slope of line AB is $\frac{-1}{8\sqrt{2}}$

Space for Rough work

10. A parallelogram is constructed with its sides parallel to the asymptotes of the hyperbola $x^2 - 2y^2 - 4x + 4y - 2 = 0$ and one of its diagonals is a chord of the same hyperbola. If the other diagonal passes through a fixed point (a, b) $a,b \in R$ then

(A) $a + b = 3$ (B) $a - b = 3$

(C) $\sum_{r=1}^{a+b} r = 6$ (D) $\sum_{r=1}^{a-b} r = 6$

11. If $m,n \in N$ such that m < n and the last three digits of 2978^m and 2978^n are same then

(A) the least value of m is 1 (B) the least value of m is 3

(C) the least value of n is 100 (D) the least value of n is 103

12. If the sum of squares of 3 real numbers a, b, c is equal to 1, then the value of expression $(4a - 3b)^2 + (5b - 4c)^2 + (3c - 5a)^2$ is always less than

(A) 50 (B) 51

(C) 52 (D) 53

13. Let $S_2 = 0$ be the mirror image of $S_1 : x^2 + y^2 - 4x - 6y + 12 = 0$ w.r.t. the line $L_1 : 10^4 x + (10^4 + 10)y + (10^4 + 20) = 0$. Let $L_2 : 2^{11}x + (2^{11} + 2^{12})y + (2^{11} + 2^{13}) = 0$ be a line. The equation of the line passing through the point of intersection of the line $L_2 = 0$ with the radical axis of $S_1 = 0, S_2 = 0$ and making equal intercepts with the coordinate axes, is

(A) $x - y - 3 = 0$ (B) $x + y + 1 = 0$

(C) $2x - 2y + 1 = 0$ (D) $2x + 2y + 3 = 0$

Space for Rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. If $g(x)=\lim_{m\to\infty}\frac{x^m f(1)+h(x)+1}{2x^m+3x+3}$ is continuous at x = 1 and $g(1)=\lim_{x\to1}(\ell n\, ex)^{\frac{2}{\ell n x}}$ then evaluate $2g(1)+2f(1)-h(1)+2$. Assume that $f(x)$ and $h(x)$ are continuous at x = 1.

2. Let $f(x)=x^3+3x+2$ and $g(x)$ be its inverse. If the area bounded by $g(x)$, x – axis and the ordinates x = – 2 and x = 6 is $\frac{p}{q}$ (where p& q are coprime), find value of $p+q-1$

3. A coin of radius 1 cm is thrown at random onto an infinite chessboard with a side length of each square as 4 cm. If the probability that the coin intersects at most one side of a square is $\frac{p}{q}$ (where p and q are coprime) then find the value of $p+q-2$

4. Consider a parabola touching x – axis at A (2, 0) and y – axis at B (0, 1). If (α,β) be the focus then $|\beta-2\alpha|$ is equal to

5. If the maximum value of $\left|\sqrt{x^4-7x^2-4x+20}-\sqrt{x^4+9x^2+16}\right|$ is P, find the value of $[P]$ (where [.] denotes GIF)

Space for Rough work